Chapter 13

Immunology

Chapter Outline

13.1 Historical Background

13.2 Organs of the Immune System

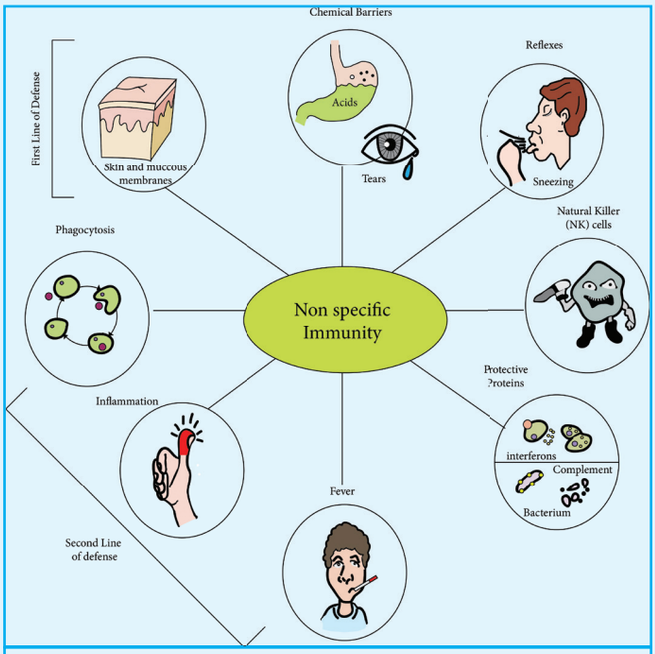
13.3 Cells of Immune System

13.4 Immunity

13.5 Antigens

13.6 Antibodies

13.7 Antigen– Antibody Reactions



Non specific immunity or Innate immunity has four types of defense barriers namely anatomical barriers, chemical barriers, phagocytic barriers and inflammatory barriers.

Learning Objectives

After studying this chapter the student will be able,

- *To gain knowledge on the history of immunology and Know the Nobel prize winners in immunology.*
- *To know the structure and functions of primary and secondary lymphoid organs.*
- *To know the cells of immune system and understand the role of granulocytes, mast cells, macrophages, dendritic cells and lymphocytes.*
- *To define immunity and Differentiate between innate immunity and acquired immunity.*
- *To understand the properties of antigen.*
- *To describe the basic structure and function of immunoglobulin (antibodies).*
- *To explain the mechanism of antigen-antibody interactions and their applications in clinical laboratory.*

13.1 Historical Background

Immunology is the study of immunity to diseases. Immunology began as a branch of Microbiology. Its origin is usually attributed to Edward Jenner who introduced variolation in 1796.